

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.60538	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	82.9971	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1550.08702	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.268095	0.268095	153.2471	0.762016	15.106667

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.60538.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 82.9971.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1550.08702.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
 - X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.312474.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.268095.
- Held-out runtime ms: 153.2471.
- Robustness dispersion across held-out structure families: 0.004803.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.400869, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.360349, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.314415, errors=[]

- X-n190-k8: feasible=True, penalized gap=0.141048, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.123794, errors=[]

Judge Note

Summary of Phase-9 CVRP Solver-Evolution Suite Results

Metric	Nearest Neighbor	Clarke-Wright Savings	Regret Insertion LS	Solver Evolution (Evolved)
Feasibility Rate	1.0	1.0	1.0	1.0
Held-out Feasible Gap	0.2734	**0.0738 (best)**	0.4664	0.2681
Held-out Runtime (ms)	42.6	83.0	1550.1	153.2
Robustness Dispersion	0.0028	0.0078	0.0869	0.0048

- ****Feasibility:**** All solvers, including the evolved one, achieved perfect feasibility (100%) on held-out problems.
- ****Objective Gap:**** The solver-evolution approach had a feasible gap (~0.268) similar to the nearest neighbor constructive (~0.273), but significantly worse than Clarke-Wright (~0.074), which had the best gap.
- ****Runtime:**** The evolved solver's runtime (~153 ms) was longer than both nearest neighbor (~43 ms) and Clarke-Wright (~83 ms), but much faster than regret insertion local search (~1550 ms).
- ****Robustness:**** The evolved solver showed low robustness dispersion (~0.0048), close to nearest neighbor and better than Clarke-Wright and regret insertion LS.

****Conclusion:****

The evolved solver maintained feasibility and robustness but did ****not**** clearly outperform the best fixed baseline (Clarke-Wright Savings), which had the lowest objective gap and competitive runtime. Hence, the solver evolution did not decisively beat the fixed baselines across held-out structure families.